

§1.2.

$$P_1. y(t) = C e^{-\int 0.5t dt} = C e^{-0.25t^2}$$

$$P_2. y(t) = C^{-\int \frac{1}{t} \sin t dt}$$

$$P_3. \mu(t) = e^{\int \frac{2t}{1+t^2} dt} = e^{\ln(1+t^2)} = 1+t^2$$

$$\Rightarrow y(t) = \frac{1}{1+t^2} \left(\int (1+t^2) \cdot \left(\frac{1}{1+t^2} \right) dt + C \right)$$

$$= \frac{1}{1+t^2} (t + C)$$

$$P_4. \mu(t) = e^{\int 1 dt} = e^t$$

$$\Rightarrow y(t) = \frac{1}{e^t} \left(\int e^t \cdot (t e^t) dt + C \right)$$

$$= e^{-t} \left(\frac{e^{2t}}{4} (2t-1) + C \right)$$

$$P_6. \mu(t) = e^{\int t^2 dt} = e^{t^3/3}$$

$$\Rightarrow y(t) = \frac{1}{e^{t^3/3}} \left(\int e^{\frac{t^3}{3}} t^2 dt + C \right)$$

$$= e^{-\frac{t^3}{3}} (e^{\frac{t^3}{3}} + C)$$

$$P_8. y(t) = y(0) e^{-\int_0^t a(s) ds} = \sqrt{5} e^{-\int_0^t \sqrt{1+s^2} ds}$$

$$P_{10}. y(t) = 0$$

$$P_{13}. \mu(t) = e^{\int 1 dt} = e^t$$

$$\Rightarrow \int_1^t \frac{d}{ds} (e^s y(s)) ds = \int_1^t \frac{e^s}{1+s^2} ds$$

$$e^t y(t) - e y(1) = \int_1^t \frac{e^s}{1+s^2} ds$$

$$y(t) = e^{-t} \left(2e + \int_1^t \frac{e^s}{1+s^2} ds \right)$$

P14. $\mu(t) = e^{\int -2t dt} = e^{-t^2}$

$$\Rightarrow \int_0^t \frac{d}{ds} (e^{-s^2} y(s)) ds = \int_0^t e^{-s^2} \cdot 1 ds$$

$$e^{-t^2} y(t) - y(0) = \int_0^t e^{-s^2} ds$$

$$y(t) = e^{t^2} \left(1 + \int_0^t e^{-s^2} ds \right)$$

P16. $(1+t^2) \frac{dy}{dt} + 4ty = t, \quad y(1) = \frac{1}{4}$

$$\Rightarrow \frac{dy}{dt} + \frac{4t}{1+t^2} y = \frac{t}{1+t^2}, \quad y(1) = \frac{1}{4}$$

Let $\mu(t) = e^{\int \frac{4t}{1+t^2} dt} = (1+t^2)^2$

So. $\int_1^t \frac{d}{ds} ((1+s^2)^2 y(s)) ds = \int_1^t (1+s^2)^2 \cdot \frac{s}{1+s^2} ds$

$$(1+t^2)^2 y(t) - 4y(1) = \int_1^t (1+s^2)s ds$$

$$(1+t^2)^2 y(t) - 1 = \left(\frac{t^2}{2} + \frac{t^4}{4} \right) - \frac{3}{4}$$

$$y(t) = \frac{1}{(1+t^2)^2} \left[\frac{t^2}{2} + \frac{t^4}{4} + \frac{1}{4} \right]$$

P18. Proof: $\frac{dy}{dt} + ay = be^{-ct}, \quad a > 0, c > 0$

$$\mu(t) = e^{\int a dt} = e^{at}$$

$$\Rightarrow y(t) = \frac{1}{e^{at}} \left(\int e^{at} \cdot (be^{-ct}) dt + C_1 \right)$$

$$= \frac{1}{e^{at}} \left(\frac{b}{a-c} e^{(a-c)t} + C_1 \right)$$

$$= \frac{b}{a-c} e^{-ct} + e^{-at} C_1$$

$$\lim_{t \rightarrow \infty} y(t) = \frac{b}{a-c} \lim_{t \rightarrow \infty} e^{-ct} + C_1 \lim_{t \rightarrow \infty} e^{-at}$$

$$= 0 \quad \text{Since } -c < 0, -a < 0$$